UK Bank Operational Resilience: Recovery Exercises and Supplier Concentration

Self-directed project on a fictional UK retail bank. Simulated internal data, real Open Banking API data as external context, FCA operational resilience framing.

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github.com/Vedant-Au/uk-bank-operational-resilience-analytics
public.tableau.com/app/profile/vedant.audichya

THE QUESTION

A bank's critical services depend on shared suppliers and have to stay within impact tolerances. Which recovery exercises should its resilience committee pilot first, and how stable is that answer when the assumptions change?

1 A warehouse first, then the question

A PostgreSQL warehouse (raw, staging, core, mart) holds 25 months of Open Banking API performance data, checksummed and reconciled, beside a seeded simulated bank: 4 services, 5 suppliers, 8 dependencies, 42 incidents. The research question was chosen only after both layers had been profiled.

2 Card authorisation breaches most, and the fix has a stable target

17 of 42 simulated episodes ran past their impact tolerance; 8 were card authorisation, which has the tightest limit at 90 minutes. Restoring it about 31% faster halves its breaches. That target holds across 50 alternative simulated histories (30.8% default, 31.3% median).

3 The exercise comparison could not pick a winner on its own

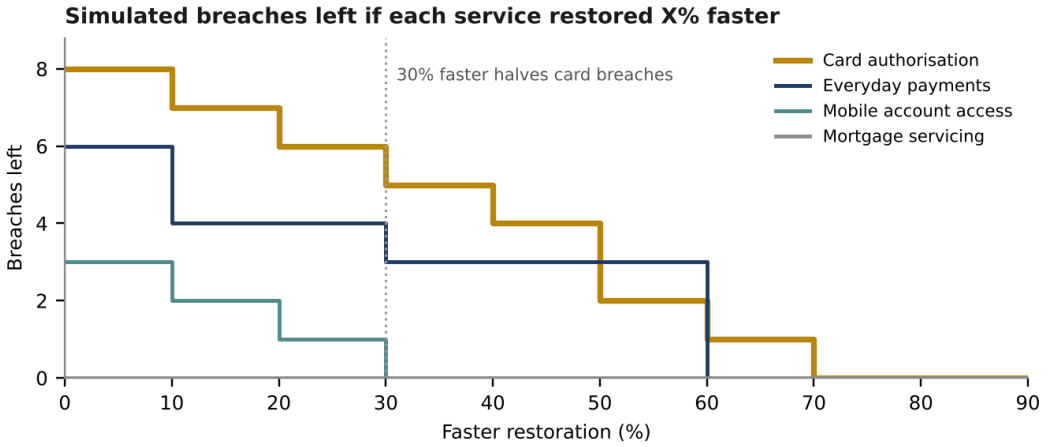
At an assumed 30% faster recovery, card and payments rehearsals each avoid a median of 2 breaches. That tie comes from the assumption, so the recovery-gap analysis was added to ask what each service needs instead of what an assumed effect delivers.

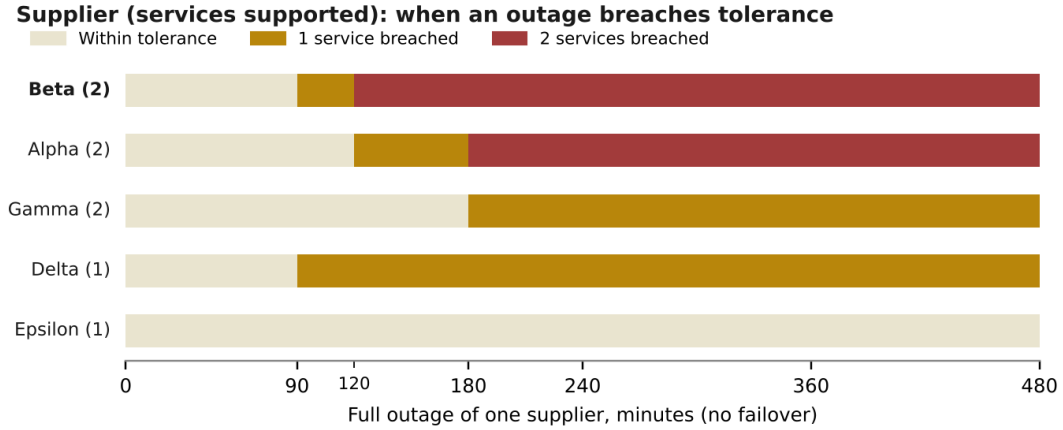
4 Provider Beta is the concentration point the history misses

Beta supports both services with the tightest tolerances and is hard to substitute: an outage over two hours breaches both. It had no shared outage in the simulated history, so replaying incidents credits a supplier exercise with almost nothing. Recommendation: rehearse card recovery first, and test a Beta outage specifically.

WHAT THIS DOES NOT SHOW

All bank incidents, suppliers, tolerances and exercise effects are simulated, so none of the figures estimate a real bank's risk. Ranges across seeds describe invented histories, not confidence intervals. The supplier stress test assumes a full outage with no failover. Open Banking data is aggregate context and is never linked to the simulated bank; it is not redistributed because it carries no open licence.





Built with PostgreSQL · Python · Excel (openpyxl) · simulation, scenario and sensitivity analysis

Delivered warehouse · simulation · scenario model · Excel dashboard · committee paper